

1. INTRODUCTION

A **health habit tracker** is an easy-to-use and efficient tool that helps users track and enhance their daily routines. It enables people to set objectives, monitor their progress, and maintain motivation to gradually establish healthier routines. Through a user-friendly interface and customised reminders, this project seeks to make habit formation more straightforward and reliable.

2. Problem Statement

Many people struggle to keep up healthy habits because they lack accountability, motivation, and a methodical way to track their progress.

Users find it difficult to see their progress in the absence of clear tracking, which results in inconsistency and a loss of motivation.

Daily obstacles and forgetfulness can interfere with the regular attention and mental effort needed to form habits.

Building long-lasting health behaviours is challenging because current approaches frequently lack personalisation and are difficult to maintain.

3. Approach and implementation

3.1 Algorithm

- **INPUT**

The program takes the following inputs from the user:

1. Water intake (glasses)
2. Sleep duration (hours)
3. Steps walked
4. Exercise minutes

The program also takes today's date automatically using date time.

- **PROCESSING**

1. Create an empty dictionary data to store daily health details.
2. Store the user's input values inside the dictionary using today's date as the key.
3. Initialise a totals dictionary to calculate weekly averages.
4. Loop through the last 7 days:
 - Generate each date
 - Check if data for that date exists in the dictionary
 - If yes, add values to totals and increase the day count
5. If at least one day's data is found:
 - Calculate the averages (total ÷ number of days)
6. Otherwise:
 - Print that no data is available.

- **OUTPUT**

1. Confirmation message:

✓ Entry saved!

2. Weekly Summary (Last 7 Days):
 - Average water intake
 - Average sleep hours
 - Average steps
 - Average exercise minutes

OR

- "No enough data for summary.

3.2 Tech Stack

- **Language** - python 3
- **Key concepts used** - Variables, Input Handling, Dictionary, Data and Time Handling, Loops, Conditional Statements, Arithmetic Operations, Data Aggregation, Output Formatting

4. Results and Analysis

The program effectively: • Gathers daily health inputs (exercise, water, sleep, and steps); • Stores today's data; • Computes the average for the previous seven days; and • Presents a weekly health summary.

The code stores daily records in a dictionary. • The loop adds available values after checking the previous seven days. • Conditions guarantee that the summary only appears when data is available. • Date time aids in accurate date computation. • Weekly habits can be quickly inferred from final averages.

CODE AND OUTPUT FOR HEALTH HABIT TRACKER

```
from datetime import datetime, timedelta

print("\n--- Daily Health Habit Tracker ---\n")

water = int(input("Water (glasses): "))
sleep = int(input("Sleep (hours): "))
steps = int(input("Steps walked: "))
exercise = int(input("Exercise minutes: "))

today = datetime.now().strftime("%Y-%m-%d")

data[today] = {
    "water": water,
    "sleep": sleep,
    "steps": steps,
    "exercise": exercise
}

print("\n✓ Entry saved!\n")

print("--- Weekly Summary (Last 7 Days) ---\n")

total = {"water": 0, "sleep": 0, "steps": 0, "exercise": 0}
days_count = 0

for i in range(7):
    date_check = (datetime.now() - timedelta(days=i)).strftime("%Y-%m-%d")

    if date_check in data:
        total["water"] += data[date_check]["water"]
        total["sleep"] += data[date_check]["sleep"]
        total["steps"] += data[date_check]["steps"]
        total["exercise"] += data[date_check]["exercise"]
        days_count += 1

if days_count > 0:
    print("Average Water      :", total["water"] // days_count)
    print("Average Sleep       :", total["sleep"] // days_count)
    print("Average Steps        :", total["steps"] // days_count)
    print("Average Exercise     :", total["exercise"] // days_count)
else:
    print("No enough data for summary.")
```

```
...
--- Daily Health Habit Tracker ---

Water (glasses): 5
Sleep (hours): 7
Steps walked: 9999
Exercise minutes: 90

✓ Entry saved!

--- Weekly Summary (Last 7 Days) ---

Average Water      : 5
Average Sleep       : 7
Average Steps       : 9999
Average Exercise    : 90
```

Key Learnings-

learnt how to use Python to accept user input. • Practiced using variables to store data. • For averages, simple arithmetic operations were used. • Recognised loops for repetitive tasks. • Outputs were printed in a clear, readable format. • Conditional statements were used to verify the data that was available.

Future Enhancements-

To ensure that data is preserved even after the program ends, add file storage. • Incorporate error handling for incorrect or absent inputs. • Produce a weekly summary and a monthly report. • Based on the user's records, add health recommendations. • Use a mobile app or graphical user interface to enhance the program's functionality.

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